

Production Planning & Scheduling Report — Q1 2026

1. Executive Summary

Q1 2026 production achieved 94.2% Overall Equipment Effectiveness (OEE) across Plant A and Plant B combined, exceeding the 92% target. Total output was 1,847,500 units against a planned volume of 1,900,000 units (97.2% plan attainment). The shortfall was primarily due to an unplanned 36-hour shutdown of CNC Line 4 in February caused by a spindle bearing failure. Preventive maintenance schedules have been updated to include quarterly vibration analysis on all CNC spindle assemblies.

2. Capacity Utilization

Plant A operated at 96.1% capacity utilization with three shifts running 6 days per week. The stamping department was the bottleneck, running at 99.3% utilization. A capital request for a new 600-ton progressive stamping press (CAPEX \$2.4M) has been submitted for Q3 installation to relieve this constraint. Plant B operated at 88.7% capacity, with the welding department showing the lowest utilization at 82.1% due to a skilled welder shortage. HR has initiated a recruitment campaign targeting AWS-certified welders.

Department	Plant	Capacity %	OEE %	Downtime Hrs	Scrap Rate %
CNC Machining	A	95.4%	93.8%	42	1.2%
Stamping	A	99.3%	96.1%	18	0.8%
Welding	A	91.2%	89.5%	67	2.1%
Assembly	A	94.8%	95.2%	31	0.5%
CNC Machining	B	92.1%	91.3%	55	1.5%
Welding	B	82.1%	84.7%	98	2.8%
Heat Treatment	B	90.5%	92.1%	44	0.3%
Finishing/Coating	B	88.3%	90.6%	52	1.1%

3. Material Requirements Planning (MRP)

MRP analysis for Q2 2026 projects a 12% increase in steel consumption driven by the new automotive transmission housing program (Part #TH-4400 series). Lead times for high-tensile steel from primary supplier Nippon Steel have increased from 8 weeks to 12 weeks due to global demand. Procurement has established a secondary supply agreement with ArcelorMittal to mitigate supply risk. Safety stock levels for critical raw materials have been increased from 2 weeks to 3 weeks coverage.

The Enterprise Resource Planning (ERP) system has been configured with updated Bill of Materials (BOM) for the TH-4400 series. Routing times have been validated through time studies conducted by Industrial Engineering. The standard cost for TH-4400 is \$47.82 per unit with a target cycle time of 4.2 minutes per piece on the CNC machining center.

4. Scheduling and Sequencing

Production scheduling follows a hybrid push-pull methodology. Make-to-stock (MTS) items use weekly frozen schedules with a 4-week planning horizon. Make-to-order (MTO) items follow customer release schedules with daily sequencing. The Advanced Planning and Scheduling (APS) module optimizes job sequencing to minimize changeover times using a genetic algorithm solver. Average changeover time has been reduced from 45 minutes to 28 minutes through SMED (Single-Minute Exchange of Die) initiatives.

5. Key Performance Indicators — Q2 Targets

KPI	Q1 Actual	Q2 Target	Stretch Goal
OEE	94.2%	95.0%	96.5%
On-Time Delivery	97.8%	98.5%	99.0%
Scrap Rate	1.3%	< 1.0%	< 0.8%
Changeover Time	28 min	25 min	20 min
Inventory Turns	8.4x	9.0x	10.0x
Customer PPM	45	< 30	< 20